

Feature Extraction for EEG

Turning a cleaned signal into numbers a classical model can learn from — band power and line length

1. Why build a classical baseline at all

Before any neural network, it's worth having a simple, fully interpretable baseline built from hand-crafted features and a classical classifier (Random Forest, SVM, etc.). Two reasons this matters, beyond just 'starting simple':

- **Sanity check:** if a complex deep learning model can't beat this simple baseline, that's a strong signal something is wrong with the deep learning approach — not evidence the problem itself is unsolvable.
- **Interpretability:** every feature here has a direct, explainable meaning a clinician could understand — unlike a learned neural network representation, which is often a black box even to the person who trained it.

2. Band power: how much energy, in which frequency range

EEG activity naturally separates into frequency bands, each loosely associated with different brain states: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–40 Hz). Seizure activity often produces a distinctive shift in how energy is distributed across these bands — which is exactly the kind of pattern a classifier can learn to recognize, if we hand it band-power numbers directly instead of the raw waveform.

How the power is actually computed

The core tool is **Welch's method**, which estimates a signal's Power Spectral Density (PSD) — literally, 'how much power exists at each frequency.' Rather than running one large Fourier transform on the whole window (which produces a noisy estimate), Welch's method:

- Splits the window into several smaller, overlapping segments
- Runs a Fourier transform on each segment separately
- Averages the results across segments

That averaging step is the entire point: it trades away a small amount of frequency precision in exchange for a much less noisy estimate — a good trade for real, noisy biological signals. Once the PSD is computed, getting the power in one specific band (say, delta) means selecting just the frequency bins that fall in that range and calculating the area under the curve over that range (numerical integration) — not just averaging the values, which would understate a band's true energy contribution.

3. Line length: a much simpler, still useful, feature

Line length is computed by taking the absolute difference between every pair of consecutive samples and summing them up. Intuitively, it measures how 'spiky' or 'jagged' a signal is: a smooth, slowly changing signal produces a small line-length value, while a chaotic, rapidly oscillating signal — the visual signature of seizure activity — produces a large one. It's inexpensive to compute and, on its own, has long been used as a surprisingly effective seizure-detection feature in the clinical EEG literature.

4. Combining features, and the real problem this surfaced

Both feature types are computed per channel, then concatenated into one flat vector per epoch: 5 band-power numbers plus 1 line-length number, per channel. This is where a real, non-obvious problem showed up in practice: if one patient has 23 channels and another has 24, their resulting feature vectors are different lengths, and no classifier can be trained on rows of inconsistent length.

The standard fix — and the one applied here — is to define one fixed, shared list of channels present across (nearly) every patient, and explicitly restrict every recording down to just those channels, in a consistent order, before extracting features. Patients recorded with a genuinely different setup are excluded from the classical baseline entirely, rather than forced into a shape that doesn't represent their real data.

Why hand-crafted features have a ceiling

Band power and line length are chosen because clinicians and researchers already know, from decades of prior work, that they correlate with seizure activity. That's their strength and their limitation at once: they can only capture patterns a human already thought to define in advance. Later modules in this roadmap — starting with a basic CNN — remove that ceiling by letting the model learn its own features directly from the raw signal, potentially discovering patterns no one explicitly specified.